

Assignment – 1

```
rushi@rushi:~$ pwd
/home/rushi
rushi@rushi:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  snap  Templates  Videos
rushi@rushi:~$ mkdir -p ~/practice/{logs,scripts,configs}
rushi@rushi:~$ cd ~/practice
rushi@rushi:~/practice$ ls
configs  logs  scripts
rushi@rushi:~/practice$ touch logs/app.log logs/error.log logs/access.log
rushi@rushi:~/practice$ tree ~/practice
/home/rushi/practice
├── configs
├── logs
│   ├── access.log
│   ├── app.log
│   └── error.log
└── scripts

4 directories, 3 files
rushi@rushi:~/practice$ printf "line1\nline2\nline3\n" > logs && head -n 2 logs/app.log
bash: logs: Is a directory
rushi@rushi:~/practice$ printf "line1\nline2\nline3\n" > logs/app.log && head -n 2 logs/app.log
line1
line2
rushi@rushi:~/practice$ pwd
/home/rushi/practice
rushi@rushi:~/practice$ ls -la
total 20
drwxrwxr-x  5 rushi rushi 4096 Aug 10 23:35 .
drwxr-x--- 17 rushi rushi 4096 Aug 10 23:35 ..
drwxrwxr-x  2 rushi rushi 4096 Aug 10 23:35 configs
drwxrwxr-x  2 rushi rushi 4096 Aug 10 23:36 logs
drwxrwxr-x  2 rushi rushi 4096 Aug 10 23:35 scripts
```

```
rushi@rushi:~/practice$ cd scripts
rushi@rushi:~/practice/scripts$ touch backup.sh deploy.sh monitor.sh
rushi@rushi:~/practice/scripts$ ls
backup.sh  deploy.sh  monitor.sh
rushi@rushi:~/practice/scripts$ cd ..
```

```
rushi@rushi:~/practice$ printf "ERROR: Database connection failed\nERROR: Server timeout\n" > logs/error.log && head -n 3 logs/error.log
ERROR: Database connection failed
ERROR: Server timeout
rushi@rushi:~/practice$
```

Assignment-2

```
rushi@rushi:~$ ls /etc/
adduser.conf      debian_version    hosts.allow       magic.mime        profile.d          subuid-
alsa              debuginfod       hosts.deny        mailcap           protocols         sudo.conf
alternatives      default          hp               mailcap.order    pulse            sudoers
anacrontab        deluser.conf     ifplugd          manpath.config   python3          sudo_logsrvd.conf
apg.conf          depmod.d         init             mime.types        python3.11       sysctl.conf
apm              dhcp            init.d           mke2fs.conf      rc0.d            sysctl.d
apparmor          dictionaries-common inputrc          ModemManager     rc1.d            systemd
apparmor.d        dpkg            inserv.conf.d   modprobe.d       rc2.d            terminfo
appport           e2scrub.conf    ipp-usb         modules          rc3.d            thermald
appstream.conf    emacs           iproute2        modules-load.d   rc4.d            thunderbird
apt              environment     iproute2        mtap             rc5.d            timezone
avahi            ethertypes      issue           nanorc           rc6.d            tmpfiles.d
bash.bashrc       fonts           issue.net       netconfig        rc5.d            ubuntu-advantage
bash_completion  fprintd.conf    kernel          netplan          rmt              ucf.conf
bash_completion.d fstab            kernel-img.conf networkd-dispatcher rpc              udev
bindresvport.blacklist fuse.conf       kernelloops.conf NetworkManager   rsyslog.conf    udisks2
binfmt.d         fwupd           krb5.conf.d     networks         rsyslog.d       ufw
bluetooth        gai.conf        ld.so.cache     newt             rygel.conf      update-manager
brlapi.key       gdb            ld.so.conf      nftables.conf   sane.d          update-motd.d
brltty           gdm3           ld.so.conf.d    nsswitch.conf   security        update-notifier
ca-certificates  geoclue        legal           openvpn          selinux         UPower
ca-certificates.conf ghostscript     libao.conf      opt             sensors3.conf   usb_modeswitch.conf
chatscripts      glvnd          libaudit.conf  os-release      sensors.d       usb_modeswitch.d
cloud            gnome          libblockdev     PackageKit      services        vim
console-setup    groff          libl3           pam.conf        sgml            vtrgb
cracklib         group          libreoffice     pam.d           shadow          vulkan
cron.d           grub.d        locale.alias    papersize       shadow-         wgetrc
cron.daily       gshadow       locale.gen      passwd          shells          wpa_supplicant
cron.hourly      gshadow-      localtime      pcmcia          skel            X11
cron.monthly     gss           logcheck        perl            snmp            xattr.conf
crontab          gss           logcheck        perl            speech-dispatcher xdg
```

```
rushi@rushi:~$ ls /var/log/
alternatives.log  btcp             dmesg            dpkg.log          installer         README           unattended-upgrades
appport.log      cloud-init.log   dmesg.0          faillog           journal          speech-dispatcher wtmp
apt              cloud-init-output.log dmesg.1.gz      fontconfig.log    kern.log         sssd
auth.log         cups            dmesg.2.gz      gdm3             lastlog         syslog
boot.log         cups-browsed    dmesg.3.gz      gpu-manager.log  openvpn         ubuntu-advantage.log
bootstrap.log    dist-upgrade     dmesg.4.gz      hp              private         ubuntu-advantage-timer.log
rushi@rushi:~$ ls /home/
rushi
rushi@rushi:~$
```

```
rushi@rushi:~$ history | awk '{print $2}' | sort | uniq -c | sort -rn | head
 95 ls
 79 cd
 28 clear
 17 mkdir
 15 rm
 12 touch
 10 pwd
 10 find
  9 sudo
  9 cat
rushi@rushi:~$
```